BrainModel: a modality-agnostic 3D foundation model for brain CT and MRI

Setup, data, preprocessing, methods and planned experiments

Working document

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Abstract

We are building a single self-supervised 3D encoder for brain CT and MRI whose frozen embeddings should serve many downstream tasks: hemorrhage and stroke detection, tumor and lesion segmentation, dementia staging, brain age and more. This document fixes the setting, the data (about 30 public sources, roughly 110,000 volumes), the shared preprocessing that places every scan in one anatomical frame, the model and training objective, the downstream evaluation protocol, and the experiments planned to finalize the model and then to post-train it with task conditioning.

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1 Motivation

Brain imaging AI is fragmented: most models are trained for one task, one modality and often one scanner. Yet CT and the many MRI contrasts image the same anatomy, and large amounts of unlabeled scans are available while labels are scarce, heterogeneous and expensive. A shared encoder pretrained without labels on both modalities could (i) give strong frozen features for new tasks with little labeled data, (ii) transfer knowledge across modalities, and (iii) provide one representation that many clinical questions can be read from. The design goal is therefore generality: an embedding that is useful across many tasks, not the best model on any single one.

2 Setting

Self-supervised pretraining. A 3D Vision Transformer is trained on unlabeled CT and MRI volumes with a joint-embedding objective (LeJEPA [1]): different crops and intensity renderings of the same volume must map to nearby embeddings, while a distribution penalty (SIGReg) keeps the embedding space isotropic and prevents collapse. Agreement is enforced both for a pooled (volume-level) embedding and token by token, and a masked-prediction term keeps local detail. Labels are never used, and downstream evaluation families are held out of pretraining entirely.

Frozen evaluation. The pretrained encoder is frozen. Every volume is embedded once (full-volume token maps), and light heads are fit on the embeddings for each labeled task: linear/ridge readouts, a small 3D convolutional probe, and a token decoder for segmentation. Token-level probes measure what the representation encodes without any task head (lesion detectability, scanner/dataset fingerprint, anatomical position). A randomly initialized encoder of the same architecture is evaluated identically and serves as the floor.

Task-conditioned post-training. After the self-supervised model is finalized (size, hyperparameters, training length, data selection and preprocessing), it is post-trained jointly on many labeled tasks, conditioned on a learned task vector that identifies the task (e.g. age regression, hemorrhage classification). The post-trained model is compared with the frozen model plus heads, on the tasks it was trained on and, crucially, on tasks held out of post-training.

Experiment records. Every evaluation writes a self-contained record (model and checkpoint provenance, data version, protocol, validation and test metrics with confidence intervals, environment, artifacts) that is ingested into one SQLite database, so models, heads, seeds and data versions are compared on identical footing.

3 Data

3.1 Sources

Table 1 lists every dataset. *Role* is **SSL** (unlabeled pretraining; the dataset may also carry downstream labels) or **held out** (never seen in pretraining; evaluation only). Counts are volumes (one volume = one 3D scan of one sequence) and subjects in the current store; they are updated by the final index build.

Table 1: Datasets. Mod.: modality and sequences in the store. Vol./Subj.: volumes and subjects.

Dataset	Description	Mod.	Vol.	Subj.	Role	Tasks
CT						
RSNA ICH [8]	Head CT studies from the RSNA 2019 Intracranial Hemorrhage Detection challenge, labeled per slice for any hemorrhage and five subtypes.	CT	25,137	25,137	SSL	`rsna_ich`

continued

Dataset	Description	Mod.	Vol.	Subj.	Role	Tasks
CQ500 [9]	Non-contrast head CTs from Indian hospitals, read by three radiologists for hemorrhage subtypes, calvarial fracture, mass effect and midline shift.	CT	422	422	held out	`cq500`
Seg-CQ500	A CQ500 subset with voxel-level intracranial hemorrhage masks.	CT	51	51	held out	`seg_cq500`, `cq500`
INSTANCE 2022	Non-contrast head CTs with intracranial hemorrhage masks from the MICCAI 2022 INSTANCE challenge.	CT	100	100	held out	`instance`
AISD [10]	Non-contrast CTs of acute ischemic stroke with infarct masks drawn with reference to follow-up diffusion MRI.	CT	398	398	held out	`aisd`
ISLES 2024	Acute stroke admission CT (non-contrast and CT angiography) with infarct masks from follow-up MRI.	NCCT, CTA	298	149	held out	`isles24`
MRI: tumors						
BraTS GLI (2023, 2024) [11]	Adult glioma, pre-operative (2023) and post-treatment (2024), four co-registered skull-stripped sequences with tumor sub-region labels.	T1, T1-Gd, T2, FLAIR	11,488	1,830	SSL	`brats2023_gli`, `brats2024_gli`
BraTS MEN 2023 [12]	Pre-operative meningioma, four skull-stripped sequences with tumor labels.	T1, T1-Gd, T2, FLAIR	4,000	944	SSL	`brats2023_men`
BraTS PED 2023 [13]	Pediatric brain tumors (mostly high-grade gliomas), four skull-stripped sequences with tumor labels.	T1, T1-Gd, T2, FLAIR	576	144	SSL	`brats2023_ped`
BraTS-GoAT	Mixed tumor types from the BraTS generalizability-across-tumors cohort, four skull-stripped sequences (via the FOMO corpus).	T1, T1-Gd, T2, FLAIR	10,444	2,611	SSL	–
MSD Brain Tumour [14]	Multi-parametric glioma MRIs from the Medical Segmentation Decathlon (Task 01); the T1 channel is used.	T1	484	484	held out	`msd`
BrainMetShare [15]	Skull-stripped multi-sequence MRI of patients with brain metastases (Stanford).	T1, T1-Gd, BRAVO, FLAIR	624	156	held out	reserved
FOMO Yale brain mets	Longitudinal, skull-stripped multi-sequence MRI of patients with brain metastases.	T1, T1-Gd, T2, FLAIR, PD	33,681	1,430	SSL	–
MRI: stroke and white-matter disease						
ATLAS [16]	T1 MRI of chronic stroke with manually traced lesion masks.	T1	1,453	1,453	held out	`atlas`
ISLES 2022 [17]	Acute and sub-acute ischemic stroke MRI with lesion masks in diffusion space.	DWI, ADC, FLAIR	750	250	held out	`isles22`
WMH 2017 [18]	FLAIR and T1 MRI from several scanners with white-matter hyperintensity masks (MICCAI 2017 challenge).	FLAIR, T1	340	170	held out	`wmh`
MRI: aging, dementia, development and healthy cohorts						
OASIS-1 [19]	Cross-sectional T1 MRI of adults aged 18–96, including very mild to mild Alzheimer's disease, with CDR.	T1	436	416	SSL	`oasis1_age`, `oasis1_cdr`
OASIS-2 [20]	Longitudinal T1 MRI of nondemented and demented older adults (60–96) with CDR at each visit.	T1	373	150	SSL	`oasis2_cdr`, `oasis2_conversion`
IXI	T1 and T2 MRI of about 600 healthy adults from three London hospitals, with demographics.	T1, T2	1,159	582	SSL	`ixi_age`
ABIDE I [21]	T1 MRI of individuals with autism spectrum disorder and typically developing controls from 17 sites.	T1	1,102	1,102	SSL	`abide_asd`
AOMIC PIOP1 [22]	T1 MRI of young adults (Amsterdam Open MRI Collection, population imaging of psychology 1).	T1	216	216	SSL	`piop1_age`
AOMIC PIOP2 [22]	As PIOP1, a second cohort.	T1	226	226	SSL	`piop2_age`

continued

Dataset	Description	Mod.	Vol.	Subj.	Role	Tasks
AOMIC ID1000 [22]	T1 MRI of about 1,000 young adults with broad demographic and psychometric data.	T1	928	928	SSL	`id1000_age`
Pixar [23]	T1 MRI of children aged 3–12 and adults from a movie-viewing fMRI study (OpenNeuro ds000228).	T1	155	155	SSL	`pixar_age`
MRI: FOMO pretraining corpus (structural sequences of selected families)						
HBN [24]	Healthy Brain Network: children and adolescents, a transdiagnostic pediatric mental-health cohort.	T1, T2, FLAIR, PD	8,481	3,843	SSL	–
NKI-Rockland [25]	Community lifespan sample from the Nathan Kline Institute.	T1, T2, FLAIR	5,951	1,326	SSL	–
CoRR [26]	Consortium for Reliability and Reproducibility: repeated scans for test–retest studies.	T1	2,745	1,498	SSL	–
BrainLat [27]	Latin American cohort with neurodegenerative diseases (AD, FTD, PD, MS) and controls.	T1, FLAIR	632	587	SSL	–
M4Raw [28]	Low-field (0.3 T) multi-contrast brain MRI.	T1, T2, FLAIR	624	208	SSL	–
Nigerian clinical	Clinical brain MRI acquired in Nigeria.	T1, T2, FLAIR	283	82	SSL	–
Used only for exclusion or not used						
BHSD	Hemorrhage segmentation labels for a subset of RSNA studies; these studies are removed from pretraining.	CT	–	–	exclusion	–
SinoCT	Reconstructed head CT series; dropped: mostly ∼36 mm slabs that cannot be placed in a whole-head frame.	CT	–	–	excluded	–

3.2 Splits and hold-outs

- **Subject-level splits.** Volumes are assigned train/validation/test (75/10/15) by subject, stratified by modality and dataset family, so all scans of a subject share one split. Splits are inherited across store rebuilds so results stay comparable; official test sets are pinned where they exist (e.g. AISD).
- **Held-out families.** CQ500/Seg-CQ500, INSTANCE, AISD, ISLES 2022/2024, ATLAS, WMH, BrainMetShare and MSD are never used for pretraining, nor are FOMO families whose source mapping overlaps them, nor the BHSD studies.
- **One pretraining volume per subject and contrast.** Duplicates (e.g. the same BraTS subject in two releases, repeat OASIS sessions) are reduced to a single volume for pretraining.
- **Pretraining overlap is recorded.** Several downstream families (OASIS, IXI, ABIDE, AOMIC, Pixar, BraTS, RSNA) are part of pretraining as unlabeled images; their test subjects are never trained on, and every evaluation record stores the fraction of the task's volumes that were in the pretraining set.

3.3 Downstream tasks

4 Preprocessing

Every scan is mapped to one stored representation: a $256 \times 256 \times 160$ grid of 1 mm voxels in a common anatomical frame, in native intensity units, with a brain mask and a record of how it got there. The steps, and why each is there:

1. **Read and orient.** Every source (NIfTI, Analyze, DICOM series) is read into one orientation convention; CT values are repaired where a converter shifted them and clamped to the valid Hounsfield range. *Why:* about 30 sources disagree on file formats, axis orders and padding values; the model must not have to learn them.

Task	Type	Data	Target	Headline metric
`cq500`	multi-label	CQ500, Seg-CQ500	ICH, IPH, IVH, SDH, SAH, calvarial fracture, mass effect, midline shift (reader majority)	macro AUC
`rsna_ich`	multi-label	RSNA	any hemorrhage, epidural, intraparenchymal, intraventricular, subarachnoid, subdural (study level)	macro AUC
`oasis1_cdr`, `oasis2_cdr`	binary	OASIS-1/2	CDR > 0	AUC
`oasis2_conversion`	binary	OASIS-2	CDR 0 at first visit, CDR > 0 later	AUC
`abide_asd`	binary	ABIDE I	autism spectrum disorder vs. control	AUC
`msd`	binary	MSD	high enhancing-tumor burden	AUC
`*_age` (6 tasks)	regression	OASIS-1, IXI, Pixar, PIOP1/2, ID1000	chronological age	R^2
`seg_cq500`, `instance`	segmentation	Seg-CQ500, INSTANCE	intracranial hemorrhage	macro Dice
`brats2023_gli`, `brats2023_ped`	segmentation	BraTS	enhancing tumor, tumor core, whole tumor	macro Dice
`brats2023_men`	segmentation	BraTS MEN	meningioma	macro Dice
`brats2024_gli`	segmentation	BraTS 2024	enhancing tumor, resection cavity	macro Dice
`atlas`, `isles22`, `isles24`, `aisd`	segmentation	stroke sets	ischemic lesion / infarct	macro Dice
`wmh`	segmentation	WMH	white-matter hyperintensities	macro Dice
`probe3d`	token probes	held-out lesion sets, all families	lesion detectability (AP lift, AUROC), modality/dataset fingerprint, position R^2	per probe

Table 2: The downstream suite: 25 entries, run for every checkpoint.

2. **Bias-field correction (raw MRI only).** N4 [5] removes the smooth intensity inhomogeneity of receive coils; releases that are already corrected are left alone. *Why:* coil shading is a scanner property, not anatomy, and would otherwise be a shortcut and a confound for intensity statistics.
3. **Registration to a template.** Each scan is centered, then fitted rigidly and affinely to the MNI152 template [6] by mutual information, which needs no intensity correspondence, so CT and every MRI contrast map to the same T1 template. Skull-stripped releases are fitted to the template's brain only. The fit runs on the CPU (SimpleITK) or on the GPU (a validated PyTorch port, about 10× faster), and the full transform is stored. *Why:* the same anatomical point should land at the same position in every scan, so the encoder's position encoding means anatomy and datasets are comparable. The fit is deliberately affine, not deformable, so individual anatomy (atrophy, ventricle size, lesions) is preserved; the stored transform recovers true physical sizes and maps outputs back onto the original scan.
4. **One resampling onto a fixed grid.** The scan is resampled once, through the fitted transform, onto $256 \times 256 \times 160$ mm at 1 mm, with anti-aliasing where the source is finer. *Why:* a single interpolation avoids compounding blur; 160 mm of depth holds the whole brain (vertex to cerebellum, 154 mm in MNI space) and is divisible by every patch depth used.
5. **Brain mask.** SynthStrip [7], a contrast- and modality-agnostic skull-stripping network, runs on the GPU; each mask is checked against the registered template brain (Dice ≥ 0.85 within the scan's field of view) with a classical fallback. *Why:* the mask provides parenchyma statistics for MRI intensity normalization, enables the skull-dropout augmentation that makes skull-on and skull-stripped scans look alike, and is a quality signal for the whole pipeline.
6. **Native intensity units.** CT stays in Hounsfield units; MRI becomes a robust z-score (median and MAD inside the brain). Windowing is not baked in; random windows are applied during training. *Why:* CT units are physical and should be kept; MRI units are arbitrary and need per-scan normalization; deferring windows lets one stored volume be seen under many renderings.
7. **Storage and index.** One compressed file per volume holds the image and its brain mask (one file read per training sample); label masks are stored alongside for downstream use. An index records subjects, splits, hold-outs, acquisition geometry (including slice thickness) and quality-control fields. *Why:* fast loading for training, and every downstream number can be traced to the exact data version.

8. **Quality control and exclusions.** Volumes that fail centering or foreground checks are not stored; SinoCT is dropped; RSNA volumes that fail the brain-mask check (mostly mis-assembled DICOM series) are excluded from pretraining but kept for evaluation. Raw data is kept as compressed archives (JPEG-LS for imaging, byte-exact for everything else) so the pipeline can be rerun.

5 Methods

5.1 Encoder

- **Architecture.** A 3D Vision Transformer (pre-norm, 12 blocks, query/key normalization), in sizes *tiny* (width 192, 3 heads) and *small* (width 384, 6 heads); *base* (768, 12) is available.
- **Tokens.** Anisotropic patches of $16 \times 16 \times 8$ mm; a full volume is a $20 \times 16 \times 16$ grid of 5,120 tokens. Patch size is variable (the patch embedding is resized, FlexiViT-style [2]) and a small network adds a function of the log patch size in mm, so the model knows the physical scale of each token.
- **Position.** Rotary position embedding [3] on the token centers in millimetres (3D, base 100 mm), so crops of any size and position share one coordinate system.
- **Air tokens** (patch center below a CT/MRI threshold) are dropped before the transformer; rows of different lengths are bucketed to limit padding.
- **Optional prefix tokens:** a CLS token and register tokens [4], without position and never dropped. With a CLS token, the pooled embedding is its output; otherwise it is the mean over patch tokens.
- **Heads used only in training:** a projector MLP for the pooled embedding, a token head (128-d) for the dense terms, and a small predictor transformer for masked prediction.

5.2 Views and augmentation

Crops are cut on the volume's own token grid, so overlapping crops share tokens exactly: two global crops ($128 \times 128 \times 64$ mm, the second inside the first), two local crops ($64 \times 64 \times 32$ mm) and two fine crops ($32 \times 32 \times 16$ mm at $8 \times 8 \times 4$ mm patches) inside a global crop, plus one crop seen under two intensity windows. Crops must contain at least 15% foreground. Appearance is re-rendered independently per crop, so that views can only be matched through anatomy, never through contrast, resolution or the presence of a skull: random intensity windows; a random monotone contrast curve with knots at tissue quantiles, inverted with probability 0.3 (as T1 and T2 order tissues differently); a smooth MRI bias field; blur along the acquisition axis up to a random slice thickness (using the recorded native thickness); isotropic blur, slow drift and noise; and, with probability 0.5, *skull dropout*, cutting a whole volume to its brain mask (only masks that passed quality control).

5.3 Objective

With $\lambda = 0.02$,

$$\mathcal{L} = (1-\lambda)\,\mathcal{L}_{\text{inv}} + \lambda\,\mathcal{L}_{\text{SIGReg}} + w_{\text{win}}\mathcal{L}_{\text{win}} + w_{\text{dense}}\big[(1-\lambda)\mathcal{L}_{\text{dense}} + \lambda\,\mathcal{L}^{\text{tok}}_{\text{SIGReg}}\big] + w_{\text{pred}}\big(\mathcal{L}_{\text{pred}} + w_{\text{pix}}\mathcal{L}_{\text{pix}}\big).$$

- $\mathcal{L}_{\text{inv}}$, $\mathcal{L}_{\text{SIGReg}}$: the LeJEPA terms on pooled embeddings: every view is pulled to the mean of the global views, and the batch of embeddings is pushed toward an isotropic Gaussian with a sketched Epps–Pulley test on random 1-D projections. $\mathcal{L}_{\text{win}}$ ties the pair of windows.
- $\mathcal{L}_{\text{dense}}$: every grid position covered by two or more views is pulled to the mean of its token embeddings across views (fine tokens averaged into their coarse position); $\mathcal{L}^{\text{tok}}_{\text{SIGReg}}$ applies SIGReg to 32 sampled token embeddings per view.
- $\mathcal{L}_{\text{pred}}$, $\mathcal{L}_{\text{pix}}$: three blocks of 3–6 tokens are hidden in each global crop; a 4-layer predictor regresses their token-head embeddings from the unmasked crop (stop-gradient target) and the $2 \times 2 \times 2$ cell means of their voxels. The weight ramps up over 2,000 steps.

5.4 Training

The *small* recipe: AdamW (lr 2.5×10^{-4}, $\beta = (0.9, 0.95)$, weight decay 0.05, none on norms, biases and prefix tokens), 2,000 warm-up steps then cosine decay, gradient clipping at 1.0, bfloat16, 30,000 steps, batch 96 volumes per GPU on 2 GPUs. Sampling is balanced across dataset families with mass $\propto n^{0.5}$, so the largest families (FOMO Yale, RSNA) do not dominate. Checkpoints record their own provenance (code version, store format and grid, index hash) so evaluations can name exactly what a model was trained on.

5.5 Frozen evaluation protocol

- **Embedding.** Each volume is encoded once, whole, under the evaluation windows (CT: brain, subdural, bone; MRI: tissue, wide), giving a 5,120-token map, a foreground mask and, for CLS models, the CLS embedding.
- **Classification and regression heads.** PCA (16 components, fitted on training tokens) followed by three readouts: per-axial-slab mean and max, a $4 \times 4 \times 4$ grid of cell means, and global mean and max, each with logistic or ridge regression; a small 3D convolutional probe on the full-width token map (seed ensemble, early stopping); and a linear head on the CLS embedding when present. Regularization is chosen on validation.
- **Segmentation.** A token U-Net decoder trained on cached token maps upsamples to one logit per voxel (cross-entropy plus soft Dice per region); training volumes are also encoded shifted off the patch grid and flipped. Dice is reported on the 64 axial slices scored by the 2D reference and on the whole volume.
- **Token probes** (`probe3d`). Lesion detection from single tokens (linear and MLP, average-precision lift over prevalence and AUROC, per held-out lesion family); fingerprinting (balanced accuracy of predicting modality and source dataset from pooled embeddings; lower is better for dataset); and location (ridge regression from a token to its normalized position, R^2, plus the participation ratio of the token covariance). Pixel-statistics and random-initialization controls are run alongside.
- **Statistics and selection.** Bootstrap 95% intervals over test volumes; for each model and task the head is chosen on validation, never on test. Results are compared across models with a suite-wide score: each task's headline metric expressed as gain over the random-initialization floor, averaged within task groups (CT classification, MRI classification, regression, segmentation, probes) and then across groups, so the many segmentation tasks do not dominate.

5.6 Task-conditioned post-training

- **Conditioning.** A learned task vector enters the encoder as an extra prefix token (the same mechanism as CLS and registers); FiLM-style modulation is a later alternative. Each task keeps its own small output head (classification, regression or segmentation decoder).
- **Task granularity.** One vector per label (each hemorrhage subtype, age, CDR) with a shared per-dataset component, so related labels can share structure; batches mix tasks.
- **Null task.** The task vector is replaced by a learned null vector for 10–20% of samples, so the model keeps a task-free embedding that can be compared like-for-like with the self-supervised model.
- **Balancing.** Temperature-based task sampling and per-task loss normalization, so RSNA (about 25,000 labeled volumes) does not drown OASIS (about 400).
- **Preserving generality.** The self-supervised loss is kept as an auxiliary term (or only adapters/LoRA are trained), and the token probes are rerun before and after post-training.

6 Planned experiments

All experiments use the v6 store, the downstream suite of Table 2, two seeds where differences are small, and the suite-wide score. A change is adopted when it improves the suite score beyond the seed spread and no task group regresses beyond its confidence interval.

	Question	Arms	Readout
E0	Floor and baseline	random-init encoder; current *small* recipe	reference rows for every comparison
E1	Prefix tokens	none (mean pooling) / 1 CLS / CLS + 4 registers	global tasks (CLS vs. token heads), dense tasks and probes (registers)
E2	Token-level SIGReg	4 vs. 32 sampled tokens per view (32 adopted)	token-probe quality, segmentation
E3	Full-volume context	+ one full-volume view per sample	global tasks; train/test context gap
E4	Mixed patch size	per-crop patch $16 \times 16 \times 8$ or $8 \times 8 \times 4$, with cross-scale view pairs	dense tasks at both scales; coarse-token quality
E5	Scale, length and data	tiny / small / base; longer schedules; sampling exponent; FOMO share; QC exclusions	suite score vs. compute
E6	Task-conditioned post-training	see Table 4	seen and held-out tasks; label efficiency; probes before/after

Table 3: Experiment plan. E0 and E1 are queued on the v6 store; E2's setting is already adopted.

Post-training comparison (E6). Roughly a third of the tasks are fixed as *held out of post-training* before any post-training run (rotated if compute allows). On seen tasks, the rungs of Table 4 separate what the representation, head capacity, supervision and conditioning each contribute. On held-out tasks, the self-supervised and post-trained backbones are compared with identical heads, and a new task vector is learned with everything else frozen (a few hundred parameters). Heads are also trained on 10% and 100% of each task's labels.

Rung	Model	Isolates
1	frozen self-supervised encoder + linear head	the representation
2	frozen self-supervised encoder + small attentive / MLP head	head capacity
3	single-task fine-tuning, one model per task	supervised ceiling per task
4	multi-task fine-tuning, no task vector	shared supervision
5	multi-task fine-tuning with task vector	conditioning

Table 4: Comparison ladder for task-conditioned post-training, on identical splits.

7 Known limitations

- **Affine size normalization.** Stored lengths and volumes are scaled by the registration (volumes by a median factor of 1.35 in our data). True sizes are recovered from the stored transform; size-sensitive endpoints must use it or condition on it.
- **RSNA DICOM assembly.** About 2% of RSNA studies mix slice thicknesses or repeat positions within a series, and gantry tilt is only removed when the affine fit succeeds. Affected volumes are excluded from pretraining; the reader will be replaced (position-based assembly with tilt correction) before the next rebuild.
- **Mixed registration engines.** Volumes are registered by SimpleITK or by the GPU port depending on the processing phase; each volume records which, and the GPU port was validated as equal or better except on heavily tilted CT.
- **Shared pretraining and evaluation images.** Several downstream families are in the unlabeled pretraining pool (Section 3.2); held-out families provide the clean test of transfer.
- **OASIS-1.** SynthStrip under-segments the atlas-registered OASIS-1 images; their masks fail quality control and are not used for skull dropout.

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